MAJOR MOON

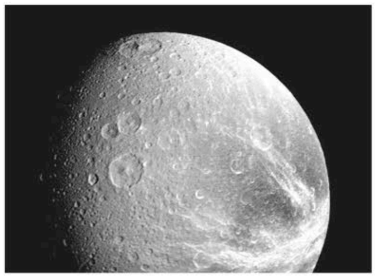
Dione

DISTANCE FROM SATURN 234,505 miles (377,400 km)

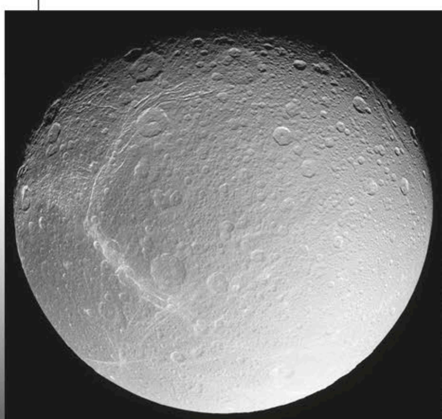
ORBITAL PERIOD 2.74 Earth days

DIAMETER 698 miles (1,123 km)

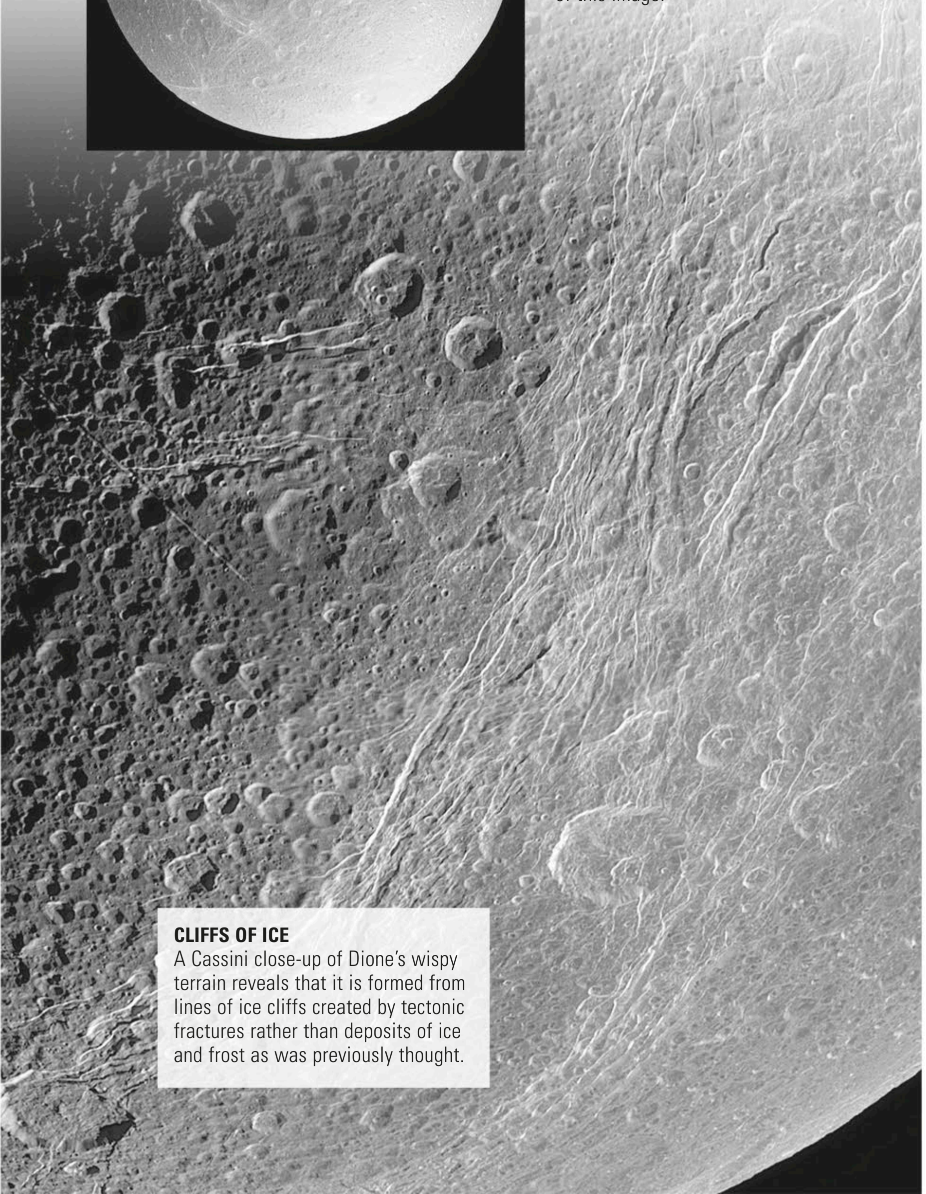
Dione is the most distant moon within Saturn's ring system, but it is not alone in the outer reaches of the E ring. Two other moons, Helene and Polydeuces, follow the same orbit; Helene is ahead of Dione by 60° and Polydeuces follows 60° behind. Helene was discovered in March 1980; Polydeuces was discovered in Cassini data some 24 years later, just after the probe arrived at Saturn. Giovanni Cassini discovered Dione in 1684, on the same day that he discovered Tethys (opposite). Dione has a higher proportion of rock in its rock–ice mix than most of the other moons (only Titan has more), so it is the second-densest of Saturn's moons. The terrain displays evidence of tectonic activity and resurfacing. There are ridges, faults, valleys, and depressions. There are also craters, which are more densely distributed in some regions than others—Dione's leading face, for example, has more than the trailing face. The largest crater is over 124 miles (200 km) across. Dione also has bright streaks on its surface. These wispy features are composed of narrow, bright, icy lines.



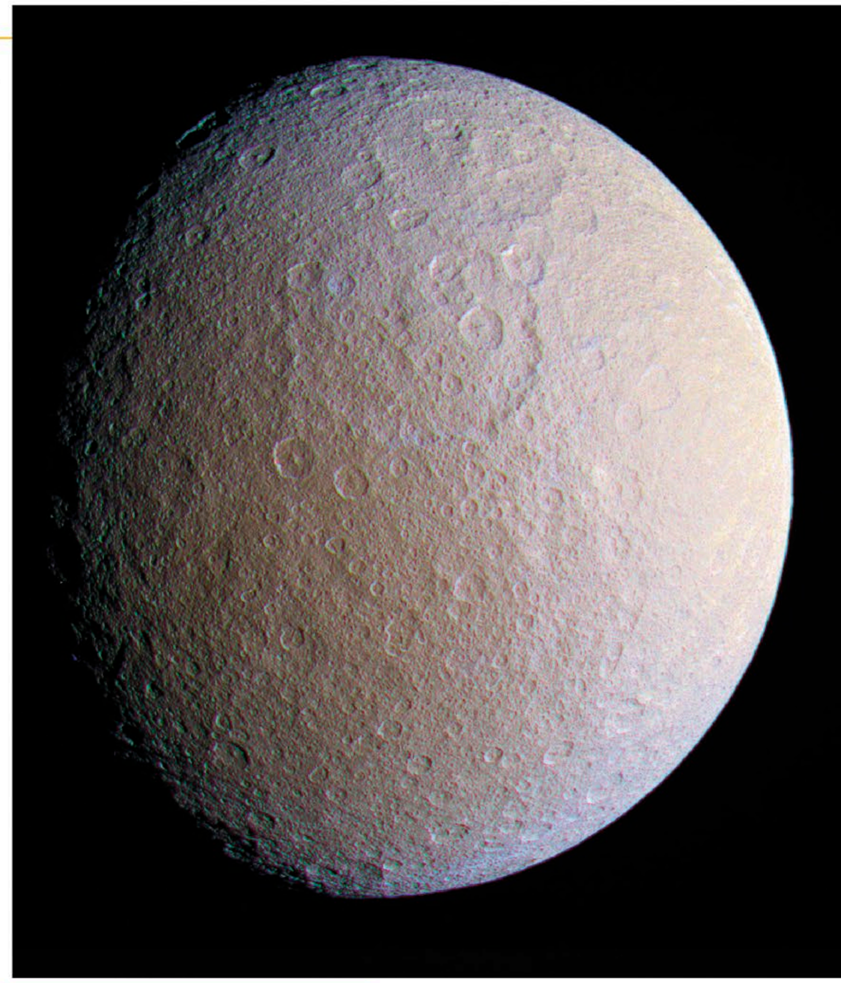
IMPACT CRATERS
The well-defined central peaks of Dione's largest craters are visible in this Voyager image. Dido Crater lies just left of center, with Romulus and Remus just above it and Aeneas Crater near the upper limb.



DIONE'S FAR SIDE
Impact craters scar the surface of the side of Dione that is permanently turned away from Saturn because it is in synchronous rotation. Areas of wispy terrain are visible on the left of this image.



CLIFFS OF ICE
A Cassini close-up of Dione's wispy terrain reveals that it is formed from lines of ice cliffs created by tectonic fractures rather than deposits of ice and frost as was previously thought.



ANCIENT SURFACE
Two large impact basins (top center) are visible in this enhanced-color image of Rhea's heavily cratered surface. The great age of these basins is indicated by the many smaller craters upon them.

MAJOR MOON

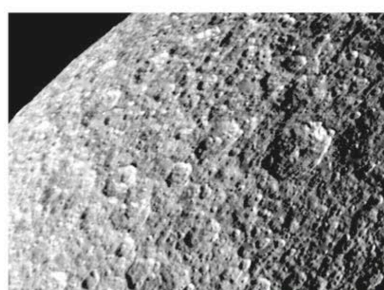
Rhea

DISTANCE FROM SATURN 327,487 miles (527,040 km)

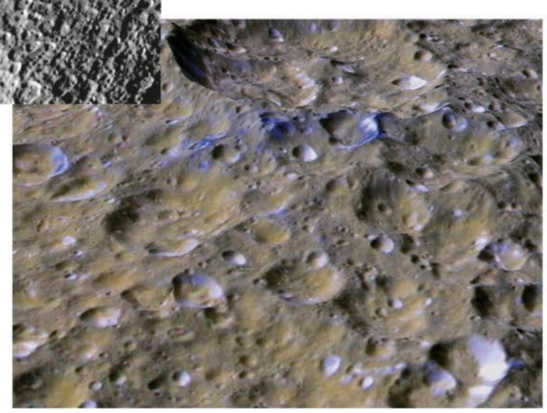
ORBITAL PERIOD 4.52 Earth days

DIAMETER 949 miles (1,527 km)

Vast sweeps of ancient cratered terrain cover large parts of Rhea. At first glance, the landscape resembles that seen on Earth's Moon, although Rhea's surface is bright ice. There is some evidence of resurfacing, although not as much as expected for such a large moon. Rhea is Saturn's second-largest moon, but other, smaller moons, such as its inner neighbors Dione and Tethys, show more resurfacing. It is thought that Rhea froze early in its history and became frigid. Its ice would then have behaved like hard rock. Rhea's craters, for example, are freshly preserved in its icy crust. The craters on other icy moons, such as Jupiter's Callisto (see p.187), have collapsed in the soft, icy crust. Rhea is the first of Saturn's moons to lie beyond the ring system. It is named after the Titan Rhea, who was the mother of Zeus in Greek mythology.



HEAVILY CRATERED
Rhea's icy surface is heavily cratered, suggesting that it dates back to the period immediately following the formation of the planets. This image shows the region around the moon's North Pole. The largest craters are several miles deep.



FRESH ICE
An enhanced-color view of the surface of Rhea shows blue patches of freshly uncovered ice. The ice is thought to have been exposed when debris in orbit around Rhea struck the surface along the equator.